CSCI 6515 – Natural Language Processing

PROJECT 2

Language Modelling and Text Classification

ADA University
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Bahruz Gurbanli
Madina Kylyshkanova

Motivation and Objectives

Goal: Build and evaluate core NLP systems.

- N-gram Modeling: Build Unigram, Bigram, and Trigram models to establish linguistic baselines.
- Smoothing Optimization: Compare Kneser-Ney and Laplace methods to fix "zero-probability" data sparsity.
- Sentiment Classification: Evaluate Naive Bayes vs. Logistic Regression using Bag-of-Words and Lexicon features.
- Boundary Detection: Use Logistic Regression (L1/L2) to distinguish terminal dots from abbreviations.
- Interactive UI: Develop a dashboard to visualize results and adjust parameters in real-time.

Dataset Overview

- Source: qz_kazakh_latn_clean_norm.jsonl (2473 documents).
- Data Split is reproducible (seed = 42).
- Train (1978 docs): Learning parameters.
- Dev (247 docs): Tuning methods and hyperparameters.
- Test (248 docs): Final generalization check.
- Vocabulary: Built from training data with min_count=2; rare words mapped to <UNK>.

Task 1 – N-Gram Language Models

- Models:
  - Unigram (1-word),
  - Bigram (2-word),
  - Trigram (3-word).
- The Metric: Perplexity (PPL).

Lower is **better**.

Key Finding: Without smoothing, higher-order models fail on new data because they cannot handle words they haven't seen in that specific order.

Model	Train PPL	Dev PPL	Test PPL	Notes
Unigram	2794.3946	2018.0583	2038.6795	finite on all splits
Bigram	29.8488	inf	inf	unseen bigrams on dev/test
Trigram	3.7229	inf	inf	more sparsity than bigram

Task 2 – Smoothing Comparison

Smoothing Goal:
Redistribute probability to "unseen" events to avoid infinite perplexity.

Methods required:
1. Laplace
2. Interpolation
3. Backoff
4. Kneser-Ney

Comparison (Test PPL) for previous dataset:

- Laplace: 5041.4 (Poor performance).
- Interpolation: 416.2.
- Backoff: 314.9.
- Kneser-Ney: 207.3 (Winner).

Method	Train PPL	Dev PPL	Test PPL
Laplace	6727.2160	11473.7357	10759.1095
Interpolation	5.2679	672.9280	532.7362
Backoff	3.7229	503.2842	398.3171
Kneser-Ney	8.1682	**335.1296**	**277.2628**

Best method (lowest dev perplexity): **Kneser-Ney**

Task 3 Sentiment DataSets

Data Processing:

- IMDB Movie Reviews dataset (50,000 reviews)
- 80% training and 20% testing
- Top 10,000 common words

Extracts **features** using:

- Bag-of-Words (CountVectorizer)
- Sentiment Lexicon

Trains four models:

- Multinomial Naive Bayes
- Bernoulli Naive Bayes
- Logistic Regression (BoW)
- Logistic Regression (Lexicon features)

Final Model Comparison:

Model	Feature	Accuracy	F1
Multinomial NB	Bag-of-Words	0.8429	0.8409
Bernoulli NB	Bag-of-Words	0.8515	0.8513
Logistic Regression	Bag-of-Words	0.8742	0.8755
Logistic Regression	Lexicon	0.7289	0.7346

Task 4 – Sentence Boundary Detection

Task: Binary classification of every period/dot in the text.
Features: Surrounding characters, casing (uppercase/lowercase), and nearby word context.
Model Comparison: Logistic Regression with L1 vs. L2 Regularization.

- Train: 31,902 (positive = 31,141)
- Dev: 4,676 (positive = 4,590)
- Test: 3,946 (positive = 3,855)

Model	Split	Accuracy	Precision	Recall	F1
L1	Train	1.0000	1.0000	1.0000	1.0000
L1	Dev	1.0000	1.0000	1.0000	1.0000
L1	Test	1.0000	1.0000	1.0000	1.0000
L2	Train	1.0000	1.0000	1.0000	1.0000
L2	Dev	0.9991	0.9991	1.0000	0.9996
L2	Test	0.9997	0.9997	1.0000	0.9999

Best model by dev F1: **L1**

Extra Task – Interactive UI

Live Demo & Visualization Content

Implementation:

- Separate UI developed using Streamlit/Python (app_p2.py).

Features:

- Live parameter adjustment for N-grams.
- Real-time Smoothing method comparison.
- Sentence detection preview (test the model on custom text).
- JSON report generation for easy data export.

Usage: Enables rapid experimentation without modifying code.

Thank You